

0/17 Questions Answered

hw6-single-prop-home-team-advantage

Student Name

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Q1 Problematic interpretation**6 Points**

Note: There are three options for homework 6, that differ by data topic, and you only need to complete one of the three options.

It is commonly thought in sports that a team has an advantage when it is a home game, meaning a game at their own stadium, field, or arena. This advantage could be due to familiarity of conditions like weather or exact field shape, support of local fans, and possibly a combination of these and more factors.

Consider the following scenario:

Maria, a student in exercise sciences and physiology, is conducting a small research study on home game advantages.

She obtains a random sample of 20 UCI sports teams and asks them whether they have more wins for home games than for away games. The results show that 10% of the teams in her sample have more wins for home games. She wants to construct a 95% confidence interval for the proportion of teams that have a higher win rate for home games than away games, among all UCI teams, so she computes the margin of error, which turns out to be 0.1315.

Q1.1**2 Points**

What is the upper bound of this CI? Enter only your final answer. Round to four decimal places (e.g. 1.23456 should be rounded to 1.2346). Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q1.2**1 Point**

What is the lower bound of this CI? Enter only your final answer. Round to four decimal places (1.23456 should be rounded to 1.2346). Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q1.3****3 Points**

Maria would like to include her results in an article that she is writing for a local newspaper but, when she computes the CI, she feels puzzled: something doesn't seem right about the lower bound. She knows you have taken a stats class, and she comes to you to ask your opinion. What is the issue with the lower bound of her CI? And what is a reason that you might suggest to Maria to explain why she faced this issue?

Save Answer

Q2 Cheating trends

11 Points

The Premier League is a professional association football (aka soccer, not American football) league in England and the highest level of the English football league system. The league is composed of 20 clubs/teams.

We want to investigate if there is a home-field advantage in football. Our sample will be from the 2019/2020 [English Premier League \(EPL\)](#), where the home team won 91 of 158 games (we will ignore the ties).

Q2.1

1 Point

First we want to understand what are we investigating and what is our data?

The population parameter of interest for this study is:

Proportion of wins in football by a home team in our data

Proportion of wins in football by a home team

Number of wins in football by a home team in our data

Number of wins in football by a home team

Save Answer

Q2.2**1 Point**

The value of the population parameter is:

91

158

 $91 / 158 = 0.54$ $158 - 91 = 67$

unknown

[Save Answer](#)**Q2.3****1 Point**

The point estimate (sample statistic) for this study is:

Proportion of wins in football by a home team in our data

Proportion of wins in football by a home team

Number of wins in football by a home team in our data

Number of wins in football by a home team

[Save Answer](#)

Q2.4**1 Point**

The value of the point estimate is:

91

158

 $91 / 158 = 0.58$ $158 - 91 = 67$

unknown

[Save Answer](#)**Q2.5****1 Point**

What are the null and alternative hypotheses to test if a majority of football games are won by the home team.

H_0 : The proportion of football games won by the home team is more than 50%; H_1 : The proportion of football games won by the home team is less than 50%

H_0 : The proportion of football games won by the home team is 50%; H_1 : The proportion of football games won by the home team is less than 50%

H_0 : The proportion of football games won by the home team is 50%; H_1 : The proportion of football games won by the home team is not 50%

H_0 : The proportion of football games won by the home team is not 50%; H_1 : The proportion of football games won by the home team is 50%

[Save Answer](#)

Q2.6**1 Point**

Let's check that the conditions are met to do a hypothesis test for a proportion. For each condition specify if it is met and explain your answer.

Condition 1: Sample data are independent?

Yes

No or unknown

Save Answer

Q2.7**1 Point**

Condition 2. More than 10 success and 10 failures in the sample?
Explain your answer.

Yes

No

Save Answer

Q2.8**1 Point**

Condition 3. Sample size is less than 10% of population? Explain your answer.

Yes

No

Save Answer

Q2.9**1 Point**

We will proceed assuming all conditions for the hypothesis test were met. If a one-sample test for proportion is performed, which of the following outcomes would lead to a rejection of the null hypothesis at a significance level of 0.05?

A p-value of 0.07

A p-value of 0.03

A p-value of 0.50

A p-value of 1.00

Save Answer

Q2.10**1 Point**

Conducting a hypothesis test for one proportion results in a p-value of .053. Which of the following is a valid conclusion from this p-value?

We accept the null hypothesis and conclude that the population proportion of football games won by a home team is 50%.

We reject the null hypothesis and conclude that the population proportion of football games won by a home team is 50%.

We reject the null hypothesis and conclude the population proportion of football games won by a home team is not 50%.

We fail to reject the null hypothesis and cannot make a conclusion about the population proportion of of football games won by a home team.

[Save Answer](#)

Q2.11**1 Point**

We actually do not just want to know if there is evidence of a home-field advantage, we want to understand the true proportion of football games won by a home team. Computation of a 95% confidence interval yields (49.9%, 65.3%). What is an appropriate interpretation of this confidence interval?

The true proportion of football games won by a home team is somewhere between 49.9% and 65.3%.

We are confident that the true proportion of football games won by a home team is somewhere between 49.9% and 65.3%.

We are 95% confident that the true proportion of football games won by a home team is somewhere between 49.9% and 65.3%.

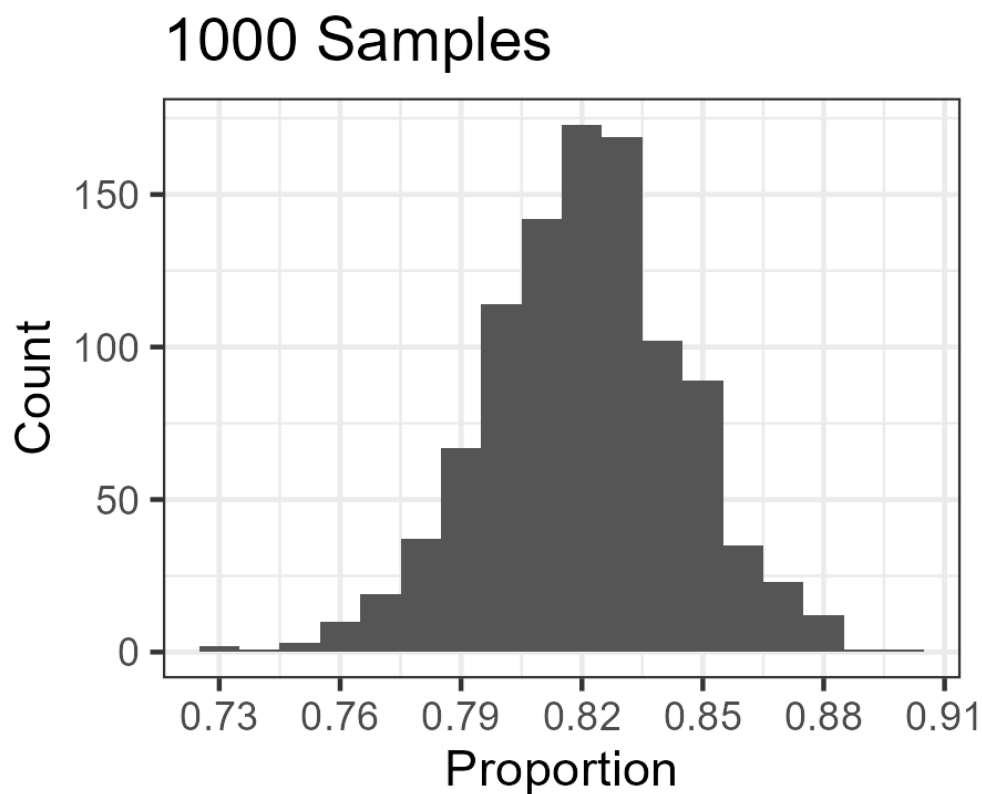
We are confident that there is a 95% probability that the true proportion of football games won by a home team is somewhere between 49.9% and 65.3%.

There is a 95% probability that the true proportion of football games won by a home team is somewhere between 49.9% and 65.3%.

Save Answer

Q3**3 Points**

Say we were able to collect 1000 samples, and for each we calculated the sample proportion, plotted below.

**Q3.1****1 Point**

What is visualized in the plot?

- sampling distribution
- distribution of sample data
- distribution of population data

Save Answer

Q3.2**2 Points**

Based on this plot, what do you think the value of the true population proportion is? Explain.

[Save Answer](#)**Q4****1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two option. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" is an insufficient response).

Please explain in detail what led you to choose this topic, home-team advantage, to investigate.

What was it about the other two options (gluten-free pasta and cheating with AI) that led you not to choose them? Please explain in detail.

[Save Answer](#)

Save All Answers

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